

Capacitive Sensor Sync

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Four capacitive sensor nodes, each an STM32 with an XBee radio, answer a single broadcast request from a MATLAB server, so all four readings belong to the same instant. A node collects 2000 ADC samples over one excitation period, discards the samples near the switching edges, and computes the plate capacitance from the averaged windows using the method in the paper.

Peripherals

Peripheral	Role
TIM1	Generates the 1 kHz plate excitation and clocks TIM2
TIM2	Triggers the ADC at 2 MHz, phase-locked to TIM1
ADC	Samples the plate voltage, 12-bit
DMA	Collects one frame of 2000 samples per period
UART	Carries the request in and the two-byte result out to the radio

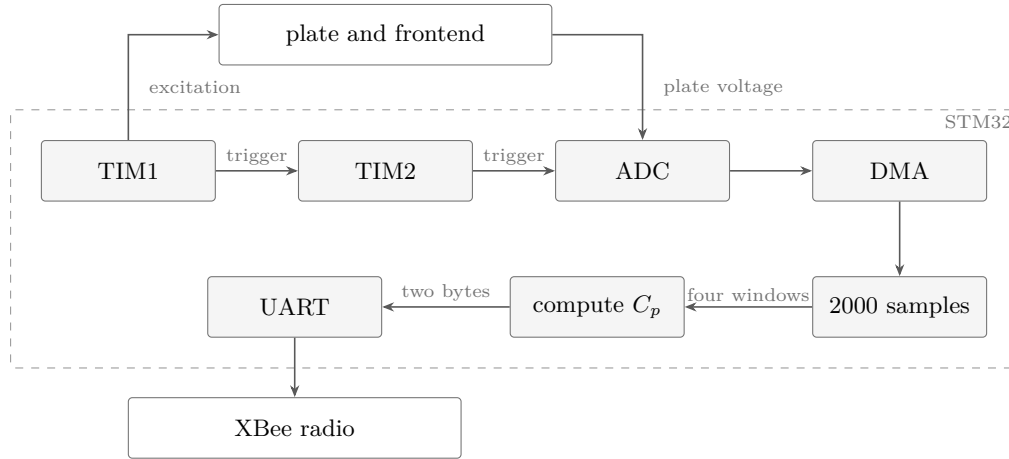
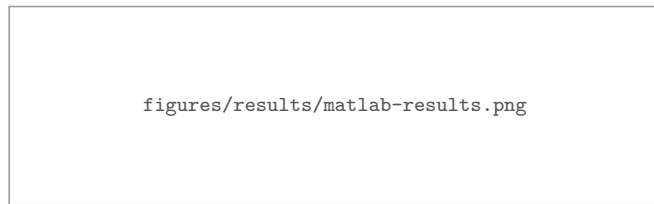
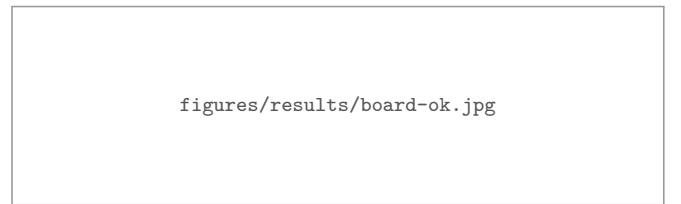


Figure 1: Acquisition path inside one node. The timers, ADC and DMA are chained in hardware; the CPU only takes over once the frame is complete.

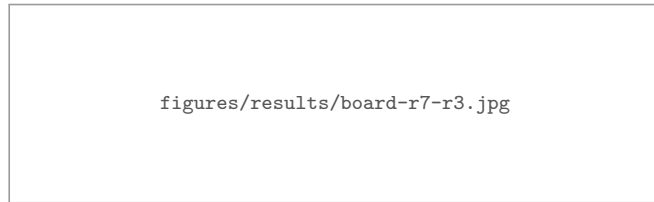
Current state



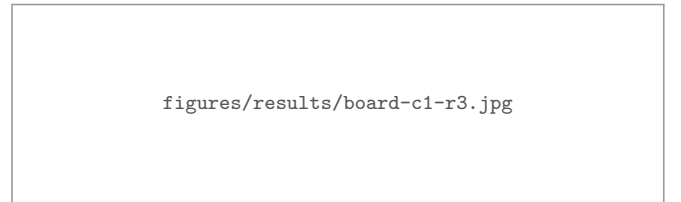
(a) MATLAB output over successive rounds.



(b) Complete board, as in the schematic.



(c) Board with R7 and R3 missing.



(d) Board with C1 and R3 missing.

Two nodes report plausible capacitance. The others read implausibly high because of the missing components shown above. Once those are repaired, the full four-channel acquisition and the modelling that follows it can proceed.